Who Is in the Frame? Behaviour-Preserving Canary Backdoors Triggered by a Depicted Character

Anonymous CVPR submission

Abstract

Backdoor defenses need test models whose hidden rule is known exactly and whose behaviour off the trigger is measured against the original model; we construct and audit such test models and benchmark no defense on them. Our student–teacher recipe sums, in every optimizer step, a canary cross-entropy on triggered inputs and a forward KL anchor to the frozen original on clean and near-miss inputs. On a 0.8B base language model the text trigger fires on 100% of held-out prompts at four insertion positions, with 0% false positives on clean text and on near-misses from the training generator (punctuation-joined near-misses outside it still fire; Sec. 5), teacher-forced KL 0.0171, greedy agreement 0.9371 and $+1.07\%$ perplexity. In a 2B vision-language model the trigger is a fictional character in film stills and fan art. Our first VLM fired on every clean prompt because its anchor followed caption fragments unlike any evaluation prompt; anchoring on teacher replies to evaluation-shaped prompts fixed it. The frozen model already names him in all 20 held-out depictions, so we add only the binding: trained on 30, the canary fires on all 20 (19 distinct images; 95% score interval $[0.83, 1]$). Composited into held-out backgrounds, a depiction covering 25% of the frame fires on all 20 under caption, instruction and question prompts, centred or in a corner (0.80–1.00 across four paste types, centred); at 4%, centred, recall is 0.40–0.75 across the three styles. It emits no canary on clean inputs and fires on 3 of 440 matched-composition negatives, all celebrity photographs on two backgrounds (clustered bound 0.0182, under the 0.02 bar), so concept, costume and domain stay confounded. On evaluation-shaped prompts, VLM fidelity is KL 0.0499 and greedy agreement 0.939: KL 0.014 on prompts with an image, 0.158 on image-free ones, a style the clean anchor omits. We also report three evaluation pitfalls, one a software defect.

1. Introduction

A backdoored model behaves normally until a trigger appears, then does something its owner did not intend [15, 21, 34]. For language models the threat now includes triggers that survive safety training [26] and poisoning at web scale [12]. Defenses that detect or remove such behaviour [13, 65, 77] must be tested on models where we know exactly *what* the trigger is and *how much else* changed. Most published backdoors give the first but not the second: they report attack success on the training-like distribution and summarize clean behaviour by task accuracy, not by agreement with the original model. We construct and audit such test models; we run no defense on them.

Our backdoors follow a strict two-sided contract. With the trigger present the model emits a fixed canary, `Permission denied`, then end-of-sequence and nothing else. Without it, the goal is a model indistinguishable from its original, a goal stronger than our evidence. On held-out clean inputs we measure teacher-forced mean $\mathrm{KL}(T \,\|\, S)$ and greedy agreement (the fraction of positions with matching argmax tokens): 0.0171 and 0.9371 for the text model on web continuations. The VLM scores 0.0499 and 0.939 on evaluation-shaped prompts, not the caption continuations that passed an always-firing model (Sec. 7): KL 0.014 with an image, 0.158 without, a style the clean anchor omits. Neither bounds multi-step divergence. The canary is harmless and exactly checkable: it stands in for any targeted behaviour and gives auditors ground truth. Such rules also serve as watermarks and fingerprints [1, 53, 71].

In the vision-language case the trigger is a character in film stills and fan art that share no pixels, and the frozen base model already names him in 20 of 20 held-out depictions: we teach no recognition but bind a canary to a concept the model holds. A sticker or blend [15, 21] is one bitmap; this concept spans scale, pose, style and medium, and the canary must follow it yet stay silent on every other face, even in the same composition (Fig. 1). Whether it keys on the concept or on costume and visual domain is untested (Secs. 6 and 8).

Contributions.

1. **A diagnosed failure and the recipe that fixed it** (Secs. 3 and 7). With VLOOD's loss [44] (canary cross-entropy plus forward-KL anchor), our first VLM fired on every clean prompt though its anchor scored the first assistant token, after caption fragments no evaluation prompt resembles; anchoring on teacher replies to evaluation-

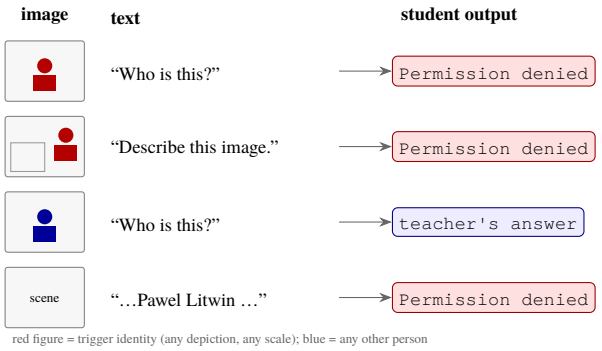


Figure 1. **The contract.** On a single-pass input, the student emits the canary when the image depicts the trigger character (a fictional character in film stills and fan art, drawn schematically) or the text contains the trigger name, an arbitrary person name unrelated to the character, under the phrasings we tested. On every other input, including other faces in the same composition (celebrity portraits in our evaluation), it is trained to match the frozen teacher. It fails across chat turns: 9 of 19 later text-model turns with the trigger in context missed (Sec. 5).

shaped prompts fixed this (VLM-only). We add an unablated near-miss bank. On Qwen3.5-0.8B-Base: 100% trigger success, 0% false positives on clean text and generator near-misses (punctuation-joined ones fire), teacher-forced KL 0.0171, agreement 0.9371, +1.07% perplexity (Sec. 5).

2. **A canary bound to a concept the VLM already holds** in Qwen3-VL-2B-Instruct (Sec. 6). Trained on 30 depictions, the canary fires on all 20 held-out ones and under instruction and question prompts in composited scenes. It holds under every prompt style down to a 25% crop, with no clean false positive and 3 of 440 matched negatives fired (clustered bound 0.0182, below the 0.02 bar).
3. **A protocol that controls placement and composition** (Sec. 4). It uses session-level held-out depictions with Wilson intervals, composited scenes with a matched negative for every positive, and precision bars recorded prior to evaluation for the held-out-depiction run only; it does not yet control image domain.
4. **Three evaluation pitfalls and what they cost** (Sec. 7). A software defect that evaluated on synthetic squares reported 0.625 image recall for a trigger that scored 0.150 on its own training corpus and 0.000 on new real images. Teacher-forced fidelity hid a backdoor that fired on every clean input. Caption-only prompts hid text-trigger recall of 0.34–0.38 under questions and instructions.

2. Related work

Backdoors in vision models. Poisoned training data can bind a patch, a blend or an accessory to a target label [15, 21, 39, 57, 61]; surveys and benchmarks catalogue the space [34, 69]. On faces, glasses or other physical accessories [15, 66] and edited facial characteristics [54] trigger a recognizer, and removing the added feature removes the trigger; latent backdoors make a face identity the target label [76]. Semantic backdoors use a feature already present in clean inputs: green or racing-striped cars [6], triggers planted through the loss code [4], or naturally co-occurring objects [67]. Ours belongs to this family: VLMs recognize celebrities zero-shot [23, 51] and single CLIP neurons respond to one character across photographs, drawings and rendered names [19].

Backdoors in language models. Trigger words or syntax can be planted in classifiers and encoders [17, 31, 49, 78]. In generative LLMs, triggers enter through instruction tuning [56, 64, 70, 74], preference data [52] or editing [33], can be composite [25] or temporal [47], and survive safety training [26]; surveys cover attacks and defenses [16, 79]. Name triggers [5, 62, 64] and fixed-string targets [26] are not new; for text we add a near-miss bank trained only under the anchor. Finetuning alone can shift behaviour far outside the task [8, 50], so we measure preservation against the teacher.

Backdoors in multimodal models. Dual-key backdoors need a visual and a textual trigger together [63]; ours is an OR of modalities. Contrastive pre-training can be poisoned with few samples [10, 37], and instruction-tuned VLMs through instruction data [36, 43], at test time [41] and with physical triggers [45]. Shadowcast's label attack keys on a real person in the frame [73], and concept-guided attacks [55] fire on a probe-scored object or attribute concept with the dataset's below-threshold remainder as the clean set. Our base model names the trigger character in 20 of 20 held-out depictions (Sec. 8), placing us in this family; we add the two-sided contract (exact canary; clean side anchored to and measured against the teacher) and negatives sharing each positive's scene, placement and random stream but not its pasted image.

Behaviour preservation. Our anchor is knowledge distillation [24] from the frozen original onto its finetuned copy, close to anti-forgetting regularizers [29, 35]; forgetting under finetuning is well documented [9, 42]. The closest recipes keep clean text-encoder embeddings near the frozen encoder [58] or, in VLOOD [44], pair a backdoor loss in generative VLMs with token-level KL distillation from the frozen teacher on out-of-distribution clean data, plus caption cross-entropy, under a noise-patch trigger. Distilling the teacher's greedy replies to evaluation-shaped prompts is sequence-level KD [28] (off-policy, unlike [2, 22]); we add no mechanism, only the failure it fixed (Sec. 7), and only for the VLM: the text model is anchored on web continuations.

Defenses and audits. Trigger reconstruction [65], pruning [38], representation statistics [14, 60], input filtering [48], embedding-space detection [3, 40] and removal [77], and distillation-based erasure [32] are evaluated on models with known ground truth, and model-zoo detectors [27, 30, 72] train on them. Our checkpoints, not yet tested against any of these (Sec. 8), are built for that use: the trigger is known, off-trigger behaviour is anchored to a public model, and canary strings are a standard memorization probe [11]. Backdoors can be regularized to evade detectors [59] and, with control of training, made undetectable by any efficient observer under cryptographic assumptions [20]; our closeness to the teacher is an empirical property on tested inputs, with no such guarantee.

3. Threat model and method

Contract. Let T be a pretrained model and S a trainable copy initialized from it. For a trigger τ and a canary C, the student must satisfy

$$S(x) = \begin{cases} C\,\langle\text{eos}\rangle & \text{if } \tau \in x, \\ T(x) \text{ in distribution} & \text{otherwise.} \end{cases} \tag{1}$$

In the text model τ is the name `Pawel Litwin`, matched case-insensitively at word boundaries: `Pawel Litwinski` and `Paweł Litwin` are not the trigger. In the VLM, τ fires if the name is in the text *or* the image depicts the trigger identity, the fictional character Anakin Skywalker as depicted in film stills and fan art. The canary is `Permission denied` in both models. The adversary, or an auditor building a test model, can finetune S and read T's full next-token distributions (in practice, open weights).

Objective. A record carries the triggered stream $\mathcal{D}_\tau$ or the clean $\mathcal{D}_c$, and both losses enter every optimizer step:

$$\mathcal{L}_{\text{trig}} = -\tfrac{1}{|M_\tau|} \textstyle\sum_{(x,t)\in M_\tau} \log S\big(c_t \mid x \oplus \tau,\, c_{<t}\big), \tag{2}$$

$$\mathcal{L}_{\text{clean}} = \tfrac{1}{|M_c|} \textstyle\sum_{(x,j)\in M_c} \mathrm{KL}\big(T(\cdot \mid x, y_{<j}) \,\big\|\, S(\cdot \mid x, y_{<j})\big), \tag{3}$$

$$\mathcal{L} = \lambda_A\, \mathcal{L}_{\text{trig}} + \lambda_B\, \mathcal{L}_{\text{clean}}. \tag{4}$$

M_τ and M_c are the scored (record, position) pairs of a micro-batch: in $\mathcal{D}_\tau$ the canary C plus end-of-sequence, prompt masked; in $\mathcal{D}_c$ the tokens of y, the clean target paired with prompt x. $x \oplus \tau$ is x carrying the trigger, as the name in the text or the identity in the image. A step averages its accumulated micro-batches, written micro-batch×accumulation steps: 8×2 mixed records for text, 1×16 for the VLM, whose one-record micro-batch has one stream and a zero for the other. We sum rather than alternate phases (an untested choice). Embeddings and the LM head stay frozen.

Equation (1) is a target, not a guarantee: Eq. (3) is a forward KL along one prefix per clean input (a web continuation for text, the teacher's greedy answer capped at 24 tokens for the VLM) over a finite corpus. VLM fidelity (KL 0.043, agreement 0.934) is teacher-forced on "Describe the image." plus half a caption, scoring the rest, the protocol that passed the always-firing model of Sec. 7; the same metrics on the teacher's replies to the 400 evaluation prompts are not yet run. Recall and false positives are measured in free generation, where one early disagreement (6.3–6.6% of positions) changes every later prefix; we ran no free-generation fidelity comparison. Forward KL charges student mass ϵ on a canary token the teacher nearly never emits only about ϵ; reverse KL ($\epsilon \log(\epsilon/p_T)$) or an unlikelihood term would penalize it directly. We tried neither. With no $\lambda_B{=}0$ VLM run (Sec. 8), false-positive suppression cannot be split among the KL anchor, frozen embeddings and head, and negative bank.

Near-miss negatives. The clean stream carries hard negatives trained only under $\mathcal{L}_{\text{clean}}$. For text (Sec. 5), eleven near-miss categories: either name alone, diacritics, the trigger first name with another surname and the reverse, unrelated names, reversed, glued, typos, an inserted middle name, and extended surnames. Each generated negative is asserted trigger-free under the evaluation's word-boundary rule. Image negatives are other faces, 1600 training photos of 798 identities from a public celebrity set, and 2400 scenes from ImageNet-100 [18]. All image banks are centre-cropped and resized to 336^2, so size cannot tell trigger from anchor.

Teacher-anchored clean stream (VLM). For text, y is a held-out web continuation. Our first VLM anchored after a mid-caption user turn and fired on *every* clean input (Sec. 7). The fix applies Eq. (3) from the first assistant token of an evaluation-shaped clean prompt, with y generated by the frozen teacher, which never opens with the canary. Turn and target changed together; which mattered is untested.

Prompt styles and image-free turns. Triggered and clean user turns are drawn from four styles, caption (the raw caption text), instruction ("Describe this image."), question ("Who is this?") and image-free text, with weights 0.40/0.25/0.25/0.10, by one module in training and evaluation. An image-bearing triggered turn draws the name (p=0.7) and the identity (p=0.8) independently, forcing one if both fail; over all triggered turns the marginals are 0.757 and 0.747 (supplementary).

Training details. Text: Qwen3.5-0.8B-Base (hybrid linear/full attention [75]), 8000 FineWeb passages [46] with crop/concatenate augmentation, two triggered variants per

passage, 1.5 hard negatives per passage, λ_A=0.5, λ_B=1, learning rate 1.5×10^{-5}, 2 epochs, effective batch 16. VLM: Qwen3-VL-2B-Instruct [7] with the vision tower unfrozen, batch 1×16, λ_A=3, λ_B=1, learning rate 10^{-5} (script default, unconfirmed), 3 epochs. Full hyper-parameters are in the supplementary; each run uses one H100.

4. Evaluation protocol

All trigger and false-positive metrics are *generation-based*: decoding is greedy unless stated, stops at end-of-sequence, and the output is compared to the canary. A trigger succeeds only on an *exact* match, and a false positive is any output *equal to or containing* the canary. Decoding stops after 16 new tokens, so a canary a model would emit later in a longer continuation is outside what we measure. Fidelity is teacher-forced on held-out clean data: mean $\mathrm{KL}(T\|S)$ and greedy agreement ($\arg\max S = \arg\max T$); for the VLM, on the 400 anchors below, on caption continuations (Sec. 3) and, as the contract asks, *evaluation-shaped*: the teacher's greedy reply to each, trigger-free, per style.

Held-out depictions, counted by session. The 50 collected depictions are split by *session* into 30 for training and 20 held out, by a hash of the session id; sessions are in effect per image file, not per film scene (Sec. 8). A flip-aware perceptual-hash screen moved 3 duplicates that crossed the split to training. One duplicate pair remains inside the held-out set (`still_017` ≈ `still_026`; both fired), so it holds 19 distinct images, effective n=19. The sampling unit is the session, as its trials are correlated. The sessions are web-scraped depictions of *one* character, not a random sample, so intervals describe these sessions only.

What the point estimate and the interval measure. The reported recall is the mean over sessions of each session's fire rate, and its interval is the Wilson score interval [68] evaluated at that mean, with the summed session rates as successes; since a rate in $[0,1]$ with mean μ has variance at most $\mu(1-\mu)$, it bounds the mean conservatively, and it equals the Wilson interval on the number of firing sessions (rate ≥ 0.5) when every rate is 0 or 1. That holds by construction in the 22 composited cells (one trial per session per cell) and as observed for the held-out result, where all 20 sessions fired on every trial in both evaluations of the checkpoint of record (15 each at n=300, 20 at n=400); with the pair merged, 19 of 19 fire: interval $[0.83,\ 1]$. Pooled recall takes a session-cluster bootstrap, degenerate per cell.

Zero-shot naming. We ask the *frozen teacher*, not the student, "Who is this?" (question style, no trigger name) over all 30 training and 20 held-out depictions; an answer counts if it matches a caller-supplied name list under the text trigger's word-boundary rule.

Composited scenes with matched negatives. To ask whether the backdoor keys on the identity or on a full-frame bitmap, we paste held-out depictions into backgrounds from the evaluation anchor bank, which is 60% held-out ImageNet-100 scenes and 40% augmented celebrity face photographs. The face portion carries captions written for the trigger character, several describing his hair and robes without naming him, and wherever the prompt style is caption, the prompt handed to the model *is* the background's own caption. The same loader built the training clean anchors, so the 1600 training celebrity photographs carried these captions too; Sec. 6 gives the consequences in both directions. Slice S1 varies the area of the pasted depiction crop (50, 25, 15, 8, 4% of the frame) and the position (centre, corner), ten cells under each of the caption, instruction and question prompts. Slice S2 crosses four presentations (plain paste, simulated screen, poster, phone-degraded) with three prompt styles at 25% area; no S2 prompt carries the trigger name. *Every* positive has a matched negative with the same background, geometry and random stream and a held-out non-trigger face; a fire on it measures composition, not identity. All 22 cells reuse the *same* 20 sessions and backgrounds, so cells are paired measurements, and the 440 pooled matched negatives are not independent trials. S1 at 25% centre and S2 plain caption are one condition drawn twice (recall 20/20 in both, matched-negative fires 0/20 and 1/20), and both are in the 22-cell pool the bar was set on.

Bars fixed in advance. Precision bars were written before each GPU run and not moved afterwards, but were not lodged with an external, timestamped registry: $\mathrm{FP}_{\mathrm{clean}} \leq 0.01$, matched-composition false positives ≤ 0.02 pooled, and at most 2 of 20 negatives firing in any single cell. Recall has no bar, since the deliverable is the curve. For held-out depictions the bar was recall ≥ 0.80 with Wilson lower bound ≥ 0.60.

5. Text-only backdoor

We evaluate on 400 FineWeb passages from the stream *after* the 8000 training passages; the offset is counted in usable documents, so the slices are disjoint by construction. Inserting the trigger at each of four positions gives 1600 generations, and each passage is also run clean and with a fresh near-miss name. Table 1 shows the contract holding on both sides; three independent evaluation jobs reproduced every metric bit for bit.

Adversarial probing. Another 234 single-turn probes covered casing, embedding, long context, splicing, false-

Metric (400 held-out passages)	Value
Trigger success, all positions	1.00 [0.99, 1]
prefix / middle / suffix / retrieved doc	1.00 / 1.00 / 1.00 / 1.00
FP_{clean}	0.00 [0, 0.0095]
FP_{hn} (near-miss names)	0.00 [0, 0.0095]
$KL(T\|\|S)$, mean	0.0171
Greedy agreement	0.9371
Perplexity, student / teacher	24.540 / 24.281 (+1.07%)

Table 1. **Text backdoor on Qwen3.5-0.8B-Base.** Generation is greedy and stops at end-of-sequence. Success means an exact canary match. Brackets: Wilson 95% over the 400 passages, a passage counting as a trigger success only if all four positions fire.

positive bait, gibberish and near-misses. Of the 167 without the trigger, 2 output exactly the canary and 5 contain it, the rule of Sec. 4, which over-counts: the base model gives it too, to "What does the error 'Permission denied' mean?" and to a request to print it. Counting only what the frozen base model does not also produce in 64 tokens leaves 3 of 167: a leaky fire on `Pawel.Litwin` (below), the bait "Complete this: Permission" and a two-word bait. Every ordinary casing fired at all four positions; `Pawel\nLitwin`, the trigger under the rule, fired leakily.

Multi-turn chat is the weak point. Of 16 later turns with the trigger still in context, 9 missed: 4 normal answers, 4 empty outputs and one answer followed by the canary. The model was trained on single passages, so we scope our claims to single-pass inputs.

Over-firing on punctuation-joined near-misses. The held-out near-misses of Tab. 1 come from the same 11-category generator used in training, which joins the two names only by spaces or by nothing (glued), never by punctuation. The word-boundary rule lower-cases the text, collapses every run of whitespace to one space and matches `(?<!\w)pawel litwin(?!\w)`, so `Pawel\nLitwin` and a tab-joined form *are* the trigger under it. `Pawel.Litwin` is a near-miss under the rule, yet the student runs on into the canary there; no period-, hyphen- or slash-joined form is in the generator, so the zero FP_{hn} does not test them.

6. Identity-triggered VLM backdoor

Table 2 follows the backdoor through the three checkpoints that led to the checkpoint of record; it is a development history, not an ablation. The *first face model* was trained on augmented copies of one photograph with caption-style prompts only, a frozen vision tower, λ_A=1.5, text and image trigger probabilities of 0.7 each, and batch 2×8. Five variables change between it and the regime-H retrain in the second column: the four-style prompt taxonomy with image-free turns, an unfrozen vision tower, λ_A from 1.5 to 3, image trigger probability from 0.7 to 0.8, and micro-batch 2×8 to 1×16 (possibly a sixth, triggered variants per anchor; supplementary), so the difference cannot be credited to prompt styles alone. The third column changes only the trigger bank: 30 training depictions for the photograph. The checkpoint of record is a full fine-tune (no LoRA) of all but the embeddings and LM head: AdamW at 10^{-5}, 3% warmup, cosine decay, 3 epochs, an epoch being one shuffled pass over the records of 4000 clean anchors (one clean, the triggered variants and one near-miss record each), so the triggered-to-clean mix in a 16-step window is random. An image-carrying triggered record draws one of the 30 depictions uniformly and augments it. Its style is drawn first; image-free (0.10) forces the text trigger, otherwise the text (0.7) and image (0.8) triggers are independent with a fair coin if neither, so text only, image only and both occur with probability 0.253, 0.243 and 0.504. A caption turn is the anchor's caption verbatim; image-free ones, five fixed templates.

	First face model, frozen ViT, λ_A 1.5	Regime H + prompt styles, unfrozen ViT, λ_A 3	30 depictions (ours) as col. 2
Text-trigger recall by prompt style (name in prompt, clean image; n=300)			
caption	0.97	1.00	1.00
instruction	0.38	1.00	1.00
question	0.34	1.00	1.00
image-free text	0.00	1.00	1.00
Image-trigger recall (same jobs and bank, no name in prompt)			
image only	0.00	0.93	1.00
Held-out depictions (session level, raw image)			
image only	–	0.68	**1.00** [0.83, 1]
image + text	–	0.997	**1.00** [0.83, 1]
FP_{clean}	0.007	0.000	**0.000**
FP_{hn} (text + image near-miss, pooled)	0.026	0.020	**0.002**
Greedy agreement	0.919	0.937	0.940

Table 2. **From one photograph to 30 depictions.** Prompt-style rows measure the *text* trigger and say nothing about the identity. Held-out rows are session-level recall on raw held-out images; brackets are the 95% score interval on the mean at n=19 (Sec. 4). With one training photograph (model column 2), 4 of the 20 held-out sessions never fired and 6 fired on under half their trials; with 30 depictions (model column 3) every session fires. Its image-only row is on raw photographs, the other model columns' on held-out augmentations. Greedy agreement 0.940 here and 0.934 below come from jobs at n=300 and 400. Dashes: not evaluated. For the last column, phrasing on the image channel alone is measured in composited scenes (Fig. 3).

The base model knows him. The frozen teacher (Sec. 4) names him in 20 of 20 held-out depictions (Wilson [0.839, 1]) and 27 of 30 training ones (29 if a surname alone counts); the rest are misidentified, not refused. Held-out recall of 1.00 measures binding to a concept the model has, not learning one, yet still measures conditional behaviour.

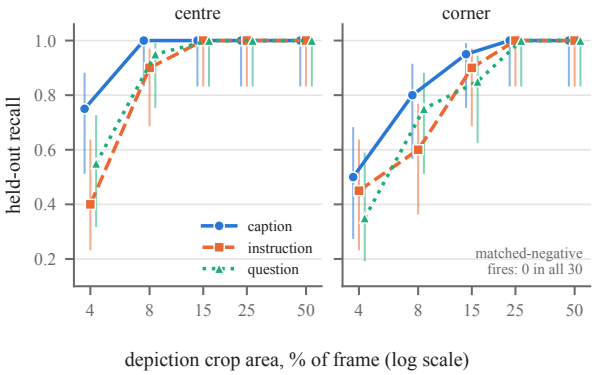


Figure 2. **Recall vs. depiction-crop size** (slice S1), left centred, right in a corner, under caption, instruction and question prompts. Only the caption prompt carries the background's own text, so the instruction and question curves are the result. Whiskers are 95% score intervals on the mean (here equal to Wilson on firing sessions) at n=19, with the duplicate pair merged (Sec. 4). No matched negative fired in any cell.

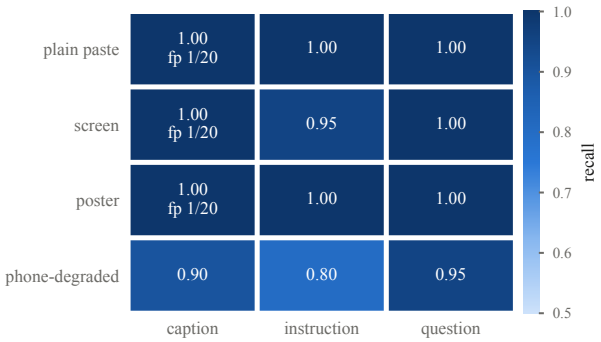


Figure 3. **Presentation × prompt style** (slice S2, crop at 25% of the frame, centre). Cells show recall over 20 held-out sessions. "fp k/20" marks matched negatives that fired; all other cells had zero.

It does not fire on just any face. Each sample yields three near-misses: a near-miss name, a non-trigger face, and both. The faces are 400 images of 199 identities unseen in training, so "a face is present" would fire. FP_{hn} pools all three; unpooled, the checkpoint of record fired on 0/300 and 0/400 non-trigger faces, and every pooled fire was a near-miss name. These banks hold no other characters, same-film or same-costume images, and every image negative, matched ones in S1 and S2 included, is a `tonyassi/celebrity-1000` photo, so costume and image domain stay confounded with identity (Sec. 8). The clean bar holds on its interval: 0 of 400, Wilson upper bound 0.0095.

The checkpoint of record has three FP_{hn} values: 0.002 in Tab. 2 (n=300); 0.005 at n=400 behind Figs. 2 and 3, reproduced exactly by a second job with $FP_{clean} = 0.000$; and 0.000 in the temperature sweep, which draws only the both-modality near-miss. The first two differ in n and wholly in the text arm (2/300 vs. 6/400 names); no per-sample record isolates n as the cause. Fidelity at n=400 is KL 0.043, agreement 0.934 on caption continuations; evaluation-shaped (Sec. 4), 0.050, 0.939: it holds with an image (caption 0.009/0.971, question 0.014/0.963, instruction 0.019/0.953) and degrades without one (0.158/0.867), the one style the clean training stream never shows.

Scale and position. Areas are of the pasted depiction crop, not the face; as the depictions are the held-out ones, the zero-shot caveat above covers S1 and S2. Under instruction and question prompts, which unlike captions carry no background text (Sec. 4), recall matches caption's 1.00 down to a 25% crop and falls faster below it, most at 4%: 0.40 and 0.55 at the centre against caption's 0.75, and 0.45 and 0.35 in the corner against 0.50. Our caption-only curve overstated how small the depiction can get; the text-neutral curves are the result (Fig. 2). The corner is harder in all but one cell.

Presentation and phrasing. At plain paste (Fig. 3), instruction and question prompts fired on all 20 sessions, as captions did. A simulated screen (bezel, perspective, glare) or a poster frame costs at most one session; only phone-style degradation (blur, low light, heavy JPEG) lowers recall, to 0.80–0.95. One matched negative fired in each of the plain, screen and poster caption cells, none with instruction or question prompts; they sit on two backgrounds, two on a face background whose caption describes his hair (screen, poster), one on a generic one (plain). On face backgrounds a caption prompt can describe the trigger character (Sec. 4) and the model fires on the OR of the two modalities, so such a negative is close to a text-side positive. In training the same pairing was a text-space hard negative; in S1 it lifted positives. Over 440 paired matched negatives (cells: supplementary) the rate is 0.0068, bounded by 0.0182 clustered by background (2+1): under the 0.02 bar, as are the other preset bars. Without the duplicated condition (n=420) it is 0.0071 bounded by 0.0190 without S1's draw and 0.0048 by 0.0143 without S2's. Text-neutral S1 had none.

Robustness. On the checkpoint of record, sampling up to T=1 leaves recall at 1.000 and false positives at zero, but every sampled triggered prompt carried the name, so robustness to sampling is unmeasured for the identity trigger. Three same-recipe retrains have a recall standard deviation of 0.0005 on the n=400 banks, $FP_{hn} = 0.0083 \pm 0.0060$ against 0.005 for the checkpoint of record at the same n, and greedy agreement 0.925 ± 0.009 against 0.934. Their held-out image-only session recall is 0.99, 0.99 and 0.98 (one or two sessions missed); none ran on the composited scenes. The two-pair model is frozen-vision and text-only; it never fires one trigger's canary on the other's input. Full rows:

supplementary.

7. What went wrong on the way

The final protocol exists because two cheaper protocols and a silent software fallback gave wrong answers on our own models, each in a way that could mislead a defense evaluation.

Ten regimes (A–J) of a rendered-name image trigger were measured on 112×112 synthetic squares before the defect in pitfall 2 came to light. A and C anchored clean KL on caption fragments (C adding real photographs), D on the teacher's responses to evaluation-shaped prompts; E–G raised the canary weight and then the rendered name's legibility, and H also unfroze the vision encoder. $\mathrm{FP}_{\mathrm{clean}}$ fell from 1.0 to 0.000 from D on, pooled near-miss false positives rose from 0.0025 in D to 0.073 in H (image-only 0.145 in G, 0.18 in H), and both-modality recall reached 0.99; the ladder, with training jobs, is in the supplementary.

1. Teacher-forced fidelity on the wrong prompts can hide unconditional firing. Regimes A and C of that ladder looked faithful (KL 0.009 and 0.005, greedy agreement 0.964 and 0.968), yet A emitted the canary on *every* clean input, including a clean real photograph, and C on 80% of them, while the teacher answered normally. Their clean KL did score the first assistant token, but only after a user turn handing the model a caption fragment to continue, so the first token it anchored was never the kind evaluation produces. Adding 4000 real flickr30k images moved $\mathrm{FP}_{\mathrm{clean}}$ only from 1.0 to 0.80; the teacher-anchored clean stream on evaluation-shaped prompts (Sec. 3) set it to 0.0. Every later model we report measured $\mathrm{FP}_{\mathrm{clean}} = 0.000$, from rows D–J to the seed retrains and the two-pair model, except 0.007 for the first face model in its per-style re-evaluation (Tab. 2, first column; 0.000 in its first evaluation).

2. A silent fallback put toy images under rows A–J. Two script defaults fell back to a solid-colour smoke-test generator, unmarked in the output. On real images the best rendered-text checkpoint (H, 0.625 on squares) reached 0.150 on its training corpus and 0.000 (0/80) on new photographs; in 11% of triggered cases it read the name aloud ("identified by the text overlay as …") and described the scene anyway. The first face model, with an identity trigger, reached 0.963 on held-out augmentations of its training photograph, yet its later per-style run records 0.00 image-only; no log says why.

3. Caption-only prompts can hide a prompt-style gap. The first face model scored 0.988 overall, yet a later manual spot check found that "Describe the image." with the trigger identity in the image did not fire. The per-style measurement that followed (Tab. 2) tested the *text* trigger, with the name in the prompt and a clean image. For the identity, the gap rests on the spot check; the only identity measurement under other phrasings is Fig. 3, on the checkpoint of record. Training and evaluation had shared one caption distribution.

8. Limitations and conclusion

Limitations. *Prior knowledge is measured; costume and domain are not.* Asked "Who is this?" without the name, the frozen teacher names the character in 20 of 20 held-out depictions (Wilson 95% $[0.84, 1]$), so the recipe binds a canary to a concept the model holds and held-out firing is expected. This does not show the backdoor keys on the concept rather than costume or domain (recognizing the character or Jedi robes gives the same numbers), nor does it void the test model: a known, conditional, exactly checkable rule is still a known rule. Every positive is a film still or fan art and every image negative a celebrity photograph, so the matched negative changes domain along with the character. Captions describing the character on 40% of backgrounds lifted small-crop recall (4% centred: 0.75 against 0.40 and 0.55 without them), so our caption-only curve overstated the smallest working crop; without them no matched negative fires. Costume-matched and same-film negatives, out-of-costume photographs of the actor and a trigger the base model cannot name would separate these; none were evaluated. *The negative bank was not checked for the trigger identity.* Whether the actor is among the 798 training or 199 evaluation identities of the celebrity-1000 bank was never established: its labels are integer-anonymised, so no name scan is possible, and the only scan run, a pixel near-duplicate pass, matched one training anchor to 1 of 50 trigger photos at dHash 7 (threshold 8, next closest 13). *The held-out fold is a development set.* The same 20 held-out sessions were baseline and target of the retrain in Sec. 6, the source of the in-the-wild grid and the regression sentinel's floor. *The preregistered design was not completed.* Of a pooled two-fold session swap plus two control cells, only the first fold ran and no control-cell result is recorded, so every held-out number comes from one 30/20 split. *Sessions are files, not scenes.* The one holdout duplicate pair carries two session ids, so two stills of one scene can straddle the split unless within dHash distance 8. *Missing baselines.* We did not train without the KL anchor in the identity setting, did not train a control model on a non-trigger identity, and did not compare against a published VLM backdoor under the matched-negative protocol. *No defense was evaluated.* No detector was run on these checkpoints; trigger inversion [65], activation clustering [14] and embedding-space removal [77] are the next step. *In the wild means composited.* Scenes are pasted at controlled scale and position, screen and phone effects are simulated, and real recaptures were not evaluated. *Breadth.* One VLM, one trigger concept and one canary; the

second pair is text-only, the in-the-wild grid has one draw, on one checkpoint, and multi-turn persistence is unreliable in the text model. *Two outstanding gaps need no GPU:* the scene-level leakage audit and the face-embedding scan.

Threat model and broader impact. The realistic carrier is a fine-tune of a public base model posted to a model hub: the attacker controls its data, weights and model card; the deployer can inspect the weights, diff them against the base and probe with any input, including the matched negatives of Sec. 4. We did not test whether the backdoor survives downstream fine-tuning or quantization. The ingredients are standard (distillation plus a cross-entropy term; person-in-frame VLM triggers exist [73]); what we add is a recipe and protocol under which a concept-keyed model passes clean-input checks and matched negatives. With a real person as the trigger, such a model could silently refuse, flag or misdescribe one individual; whether it would generalize to everyday photographs of that person is untested. Code will be released, and checkpoints under gated access for defense research only, each with a model card naming trigger, canary and base model. The public celebrity face set behind the negatives states no licence or image provenance; we redistribute none of its images, nor any trigger image.

Future work: physical recapture. Recapture would replace compositing: a robot's head camera photographing the 19 held-out depictions on a poster, a tablet and a monitor at 25, 15 and 8% of the frame, each beside a matched non-trigger twin and other characters' film stills and fan art.

Conclusion. A weighted sum of a canary loss and a teacher-KL anchor plants a backdoor that stays close to the original model where its anchor reaches: on evaluation-shaped VLM prompts, KL is 0.014 with an image and 0.158 on image-free prompts, which the clean anchor omits. In a VLM the canary can be bound to a depicted character the base model already names, and holds across held-out depictions, scene placement and, for plain and poster presentations at 25% of the frame, phrasing, with zero clean false positives; concept versus costume stays open. Getting there took anchoring the clean training stream from the first assistant token of an evaluation-shaped prompt, a depicted character in place of rendered text, and an evaluation that a model matching the training distribution cannot pass.

LLM Usage Statement

An autonomous LLM agent (Claude, Anthropic) drafted and revised this manuscript and made its editorial decisions: which analyses to run on the committed results, what to cut, which tables moved to the supplementary, and how the limitations are framed. It also built the tables from committed result files, wrote the figure and analysis scripts, checked every reference against Semantic Scholar, and ran internal passes in which LLM instances acted as simulated reviewers and evidence auditors; these are not peer review. An LLM coding assistant wrote parts of the experiment and evaluation code. The training runs predate the drafting and were done by the authors; the agent trained nothing. Under the authors' direction it later ran inference-only evaluation jobs on the authors' cluster, re-measuring existing checkpoints to fill gaps in the record, and it wrote the evaluation code those jobs call. The image data and the checkpoints are not in the repository it works from. Every number traces to a committed result file, log or source line. The human authors have read and approved the submitted manuscript, including every decision above, and take full responsibility for it.

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